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PLACEMENT SUCCESS INTELLIGENCE ANALYSIS

TECHNOLOGIES USED: POWER BI | PYTHON (PANDAS) | DATA VISUALIZATION | CANVA | KPI ANALYSIS

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PROJECT OVERVIEW :

This report presents a comprehensive analysis of student placement performance across five engineering departments. Using Power BI, data-cleaning, KPI analysis, this study examined 5,000 students to identify key drivers of placement success and roadblocks that lead to minimal errors in career outcomes.

The analysis reveals a placement rate of 36.24%, with CSE and IT departments **leading performance**. **Critical components** such as CGPA, projects, and certifications were identified as significant components. **roadblocks**, including academic backlogs and low coding skills. Key recommendations have been introduced to improve the placement performance amongst the students.

P**rimary goal:** Identifying the academic and skill factors impacting student placements.

Strategic Plan to improve: Analyse the (3,188) unplaced students to design a training budget program based on departments and skill defects.

METHODOLOGIES:

1. Data Collection and Exploration:

- **How the data was collected :**

  The dataset was collected from Kaggle datasets, containing campus placement details, academic scores etc for 5000 students. . It was downloaded into a desktop folder and then into GitHub which was later loaded into pandas data frame for analysis.

A). DATA EXPLORATION

```
!git clone https://github.com/rishitha-c-k/Placement-success-dashboard-.git

Cloning into 'Placement-success-dashboard-'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), 638.02 KiB | 3.63 MiB/s, done.
```

```
!ls

Placement-success-dashboard-
```

```
!ls Placement-success-dashboard-/

test.csv.csv  train.csv
```

```
df = pd.read_csv('/content/Placement-success-dashboard-/train.csv')
df.head()
```

	Student_ID	Age	Gender	Degree	Branch	CGPA	Internships	Projects	Coding_Skills	Communication_Skills	Aptitude_Test_Score	Soft_Skills_Rating	Certificat
0	1048	22	Female	B.Tech	ECE	6.29	0	3	4	6	51	5	
1	37820	20	Female	BCA	ECE	6.05	1	4	6	8	59	8	
2	49668	22	Male	MCA	ME	7.22	1	4	6	6	58	6	
3	19467	22	Male	MCA	ME	7.78	2	4	6	6	90	4	
4	23094	20	Female	B.Tech	ME	7.63	1	4	6	5	79	6	

Next steps: Generate code with df New interactive sheet

fig1. Loading of dataset into pandas dataframe

- **To check databases:** Shape(), info(), describe()

```
df.shape
```

```
(45000, 15)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 45000 entries, 0 to 44999
Data columns (total 15 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Student_ID            45000 non-null  int64
 1   Age                   45000 non-null  int64
 2   Gender                45000 non-null  object
 3   Degree                45000 non-null  object
 4   Branch                45000 non-null  object
 5   CGPA                  45000 non-null  float64
 6   Internships           45000 non-null  int64
 7   Projects              45000 non-null  int64
 8   Coding_Skills         45000 non-null  int64
 9   Communication_Skills  45000 non-null  int64
 10  Aptitude_Test_Score   45000 non-null  int64
 11  Soft_Skills_Rating    45000 non-null  int64
 12  Certifications        45000 non-null  int64
 13  Backlogs              45000 non-null  int64
 14  Placement_Status      45000 non-null  object
dtypes: float64(1), int64(10), object(4)
memory usage: 5.1+ MB
```

```
df.describe()
```

	Student_ID	Age	CGPA	Internships	Projects	Coding_Skills	Communication_Skills	Aptitude_Test_Score	Soft_Skills_Rating	Certifi
count	45000.000000	45000.000000	45000.000000	45000.000000	45000.000000	45000.000000	45000.000000	45000.000000	45000.000000	4500C
mean	24977.962600	20.999333	7.002290	0.774089	3.734222	5.691800	5.501644	69.385356	5.501644	1
std	14425.605704	1.995071	0.993855	0.844750	0.923738	1.994674	1.515374	13.909710	1.238722	C
min	1.000000	18.000000	4.500000	0.000000	1.000000	1.000000	1.000000	35.000000	1.000000	C
25%	12509.750000	19.000000	6.320000	0.000000	3.000000	4.000000	4.000000	60.000000	5.000000	1
50%	24957.500000	21.000000	7.000000	1.000000	4.000000	6.000000	6.000000	69.000000	5.000000	2
75%	37475.250000	23.000000	7.670000	1.000000	4.000000	7.000000	7.000000	79.000000	6.000000	2
max	50000.000000	24.000000	9.800000	3.000000	6.000000	10.000000	10.000000	100.000000	10.000000	3

- **To view first and last rows:** head(), tail()

```
df = pd.read_csv('/content/Placement-success-dashboard-/train.csv')
display(df.head())
```

	Student_ID	Age	Gender	Degree	Branch	CGPA	Internships	Projects	Coding_Skills	Communication_Skills	Aptitude_Test_Score	Soft_Skills_Rating	Certifications	Backlogs	Placement_Status
0	1048	22	Female	B.Tech	ECE	6.29	0	3	4	6	51	5	1	3	Not Placed
1	37820	20	Female	BCA	ECE	6.05	1	4	6	8	59	8	2	1	Not Placed
2	49668	22	Male	MCA	ME	7.22	1	4	6	6	58	6	2	2	Not Placed
3	19467	22	Male	MCA	ME	7.78	2	4	6	6	90	4	2	0	Placed
4	23094	20	Female	B.Tech	ME	7.63	1	4	6	5	79	6	2	0	Placed

```
df.tail(8)
```

	Student_ID	Age	Gender	Degree	Branch	CGPA	Internships	Projects	Coding_Skills	Communication_Skills	Aptitude_Test_Score	Soft_Skills_Rating	Certifications	Backlogs	Placement_Status
44992	49628	22	Female	BCA	IT	6.26	0	3	4	10	53	7	1	1	Not Placed
44993	41608	24	Male	B.Sc	ECE	7.13	1	4	5	6	94	4	2	0	Placed
44994	17135	23	Male	BCA	ECE	6.30	0	4	7	4	65	5	2	3	Not Placed
44995	9362	20	Male	MCA	Civil	7.84	2	5	9	7	68	6	3	1	Placed
44996	8940	19	Female	B.Sc	ME	8.28	1	4	6	5	74	5	2	0	Placed
44997	13097	20	Female	B.Sc	Civil	8.88	0	4	4	5	83	4	2	0	Not Placed
44998	12958	24	Female	MCA	ECE	5.90	1	3	6	6	60	2	2	1	Not Placed
44999	9565	23	Female	B.Tech	ME	6.28	0	4	6	6	64	7	2	3	Not Placed

- **To find unique values:** unique(), nunique(), value_counts()

```
df['Coding_Skills'].unique()
```

```
array([ 4,  6,  5,  3,  7,  1,  9,  8,  2, 10])
```

```
df['Projects'].unique()
```

```
array([3, 4, 2, 5, 6, 1])
```

```
df['Branch'].nunique()
```

```
5
```

```
df['Projects'].nunique()
```

```
6
```

value_counts --> this shows how many times each unique value has repeated.

```
df['Placement_Status'].value_counts()
```

Placement_Status	count
Not Placed	28688
Placed	16312

dtype: int64

- **To measure statistics:** min(), max(), mean()

```
df['Aptitude_Test_Score'].max()
```

```
100
```

```
df['Backlogs'].min()
```

```
0
```

```
df['CGPA'].mean()
```

```
np.float64(7.0022902222222205)
```

. **Group by branch and gender:** group.by().mean()

To see average cgpa of each branch

```
avg_cgpa= df.groupby('Branch')['CGPA'].mean()
display(avg_cgpa)
```

Branch	CGPA
CSE	7.006738
Civil	7.014775
ECE	7.005899
IT	6.994024
ME	6.990065

dtype: float64

to see average cgpa of each gender

```
avg_cgpa_gender=df.groupby('Gender')['CGPA'].mean()
display(avg_cgpa_gender)
```

Gender	CGPA
Female	7.001018
Male	7.003576

dtype: float64

- **Accessing specific rows and columns:** loc(), type()

```
df.loc[120]
```

	120
Student_ID	29131
Age	20
Gender	Female
Degree	BCA
Branch	ME
CGPA	6.38
Internships	0
Projects	3
Coding_Skills	4
Communication_Skills	5
Aptitude_Test_Score	61
Soft_Skills_Rating	5
Certifications	1
Backlogs	3
Placement_Status	Not Placed

dtype: object

```
df.loc[120,'Soft_Skills_Rating']
```

```
np.int64(5)
```

```
type(df)
```

```
pandas.core.frame.DataFrame
def __init__(data=None, index: Axes | None=None, columns: Axes | None=None, dtype: Dtype |
None=None, copy: bool | None=None) -> None
    ``copy=False`` will ensure that these inputs are not copied.

    .. versionchanged:: 1.3.0

See Also
--------
```

```
type(df['Age'])
```

```
pandas.core.series.Series
def __init__(data=None, index=None, dtype: Dtype | None=None, name=None, copy: bool | None=None,
fastpath: bool | lib.NoDefault=lib.no_default) -> None
    Contains data stored in Series. If data is a dict, argument order is
    maintained.
index : array-like or Index (1d)
    Values must be hashable and have the same length as `data`.
    Non-unique index values are allowed. Will default to
    RangeIndex (0, 1, 2, ..., n) if not provided. If data is dict-like
    and index is None, then the keys in the data are used as the index. If the
```

B). DATA CLEANING

- **To check null values and count null per column:** isnull(), isnull().sum()

isnull() --> gives all the null values present in dataset

```
df.isnull()
```

	Student_ID	Age	Gender	Branch	CGPA	Internships	Projects	Coding_Skills	Communication_Skills	Aptitude_Test_Score	Soft_Skills_Rating	Certifications	Backlogs	Placement_Status
0	False	False	False	False	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
44995	False	False	False	False	False	False	False	False	False	False	False	False	False	False
44996	False	False	False	False	False	False	False	False	False	False	False	False	False	False
44997	False	False	False	False	False	False	False	False	False	False	False	False	False	False
44998	False	False	False	False	False	False	False	False	False	False	False	False	False	False
44999	False	False	False	False	False	False	False	False	False	False	False	False	False	False

45000 rows × 14 columns

```
df.isnull().sum()
```

	0
Student_ID	0
Age	0
Gender	0
Branch	0
CGPA	0
Internships	0
Projects	0
Coding_Skills	0
Communication_Skills	0
Aptitude_Test_Score	0
Soft_Skills_Rating	0
Certifications	0
Backlogs	0
Placement_Status	0

dtype: int64

- **Fill null values :** fillna()

```
df['Coding_Skills'].fillna(0, inplace=False)
df.head(8)
```

	Student_ID	Age	Gender	Branch	CGPA	Internships	Projects	Coding_Skills	Communication_Skills	Aptitude_Test_Score	Soft_Skills_Rating	Certifications	Backlogs	Placement_Statu
0	1048	22	Female	ECE	6.29	0	3	4	6	51	5	1	3	Not Place
1	37820	20	Female	ECE	6.05	1	4	6	8	59	8	2	1	Not Place
2	49668	22	Male	ME	7.22	1	4	6	6	58	6	2	2	Not Place
3	19467	22	Male	ME	7.78	2	4	6	6	90	4	2	0	Place
4	23094	20	Female	ME	7.63	1	4	6	5	79	6	2	0	Place
5	8710	20	Male	ECE	7.99	1	4	5	7	84	6	2	0	Place
6	24363	21	Male	ECE	7.50	1	4	6	4	71	8	2	1	Not Place
7	27448	19	Male	ME	8.00	0	4	4	5	74	3	1	0	Not Place

- **Drop null values:** dropna()

```
df.dropna().head(8)
```

	Student_ID	Age	Gender	Branch	CGPA	Internships	Projects	Coding_Skills	Communication_Skills	Aptitude_Test_Score	Soft_Skills_Rating	Certifications	Backlogs	Placement_Status
0	1048	22	Female	ECE	6.29	0	3	4	6	51	5	1	3	Not Placed
1	37820	20	Female	ECE	6.05	1	4	6	8	59	8	2	1	Not Placed
2	49668	22	Male	ME	7.22	1	4	6	6	58	6	2	2	Not Placed
3	19467	22	Male	ME	7.78	2	4	6	6	90	4	2	0	Placed
4	23094	20	Female	ME	7.63	1	4	6	5	79	6	2	0	Placed
5	8710	20	Male	ECE	7.99	1	4	5	7	84	6	2	0	Placed
6	24363	21	Male	ECE	7.50	1	4	6	4	71	8	2	1	Not Placed
7	27448	19	Male	ME	8.00	0	4	4	5	74	3	1	0	Not Placed

- **To check Duplicated and drop.duplicated:** duplicated(), drop_duplicated()

```
df.duplicated()
```

	0
0	False
1	False
2	False
3	False
4	False
...	...
44995	False
44996	False
44997	False
44998	False
44999	False

45000 rows × 1 columns

dtype: bool

```
df.drop_duplicates()
```

	Student_ID	Age	Gender	Branch	CGPA	Internships	Projects	Coding_Skills	Communication_Skills	Aptitude_Test_Score	Soft_Skills_Rating	Certifications	Backlogs	Placement_Status
0	1048	22	Female	ECE	6.29	0	3	4	6	51	5	1	3	Not Placed
1	37820	20	Female	ECE	6.05	1	4	6	8	59	8	2	1	Not Placed
2	49668	22	Male	ME	7.22	1	4	6	6	58	6	2	2	Not Placed
3	19467	22	Male	ME	7.78	2	4	6	6	90	4	2	0	Placed
4	23094	20	Female	ME	7.63	1	4	6	5	79	6	2	0	Placed
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
44995	9362	20	Male	Civil	7.84	2	5	9	7	68	6	3	1	Placed
44996	8940	19	Female	ME	8.28	1	4	6	5	74	5	2	0	Placed
44997	13097	20	Female	Civil	8.88	0	4	4	5	83	4	2	0	Not Placed
44998	12958	24	Female	ECE	5.90	1	3	6	6	60	2	2	1	Not Placed
44999	9565	23	Female	ME	6.28	0	4	6	6	64	7	2	3	Not Placed

45000 rows × 14 columns

To drop a column: drop(columns)

to drop a column(degree) and display the entire dataset

```
df.drop(columns=["Degree"], inplace=True, errors='ignore')
display(df)
```

	Student_ID	Age	Gender	Branch	CGPA	Internships	Projects	Coding_Skills	Communication_Skills	Aptitude_Test_Score	Soft_Skills_Rating	Certifications	Backlogs	Placement_Status
0	1048	22	Female	ECE	6.29	0	3	4	6	51	5	1	3	Not Placed
1	37820	20	Female	ECE	6.05	1	4	6	8	59	8	2	1	Not Placed
2	49668	22	Male	ME	7.22	1	4	6	6	58	6	2	2	Not Placed
3	19467	22	Male	ME	7.78	2	4	6	6	90	4	2	0	Placed
4	23094	20	Female	ME	7.63	1	4	6	5	79	6	2	0	Placed
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
44995	9362	20	Male	Civil	7.84	2	5	9	7	68	6	3	1	Placed
44996	8940	19	Female	ME	8.28	1	4	6	5	74	5	2	0	Placed
44997	13097	20	Female	Civil	8.88	0	4	4	5	83	4	2	0	Not Placed
44998	12958	24	Female	ECE	5.90	1	3	6	6	60	2	2	1	Not Placed
44999	9565	23	Female	ME	6.28	0	4	6	6	64	7	2	3	Not Placed

45000 rows × 14 columns

Next steps: Generate code with df | New interactive sheet

To filter and sort values: sort_values()

```python
df[['CGPA', 'Placement_Status']].sort_values(by='CGPA', ascending=False)
```

	CGPA	Placement_Status
10100	9.8	Placed
22693	9.8	Placed
22459	9.8	Not Placed
22409	9.8	Not Placed
10242	9.8	Not Placed
...	...	...
13027	4.5	Not Placed
43593	4.5	Not Placed
42787	4.5	Not Placed
33031	4.5	Not Placed
38894	4.5	Not Placed

45000 rows × 2 columns

getting internship column values in descending order acending=True --> ascending order, ascending=False --> Descendind order

```python
df['Internships'].sort_values(ascending=False).head(6)
```

	Internships
29316	3
29324	3
3769	3
29217	3
29242	3
3787	3

dtype: int64

2. Data Visualization:

- **How is data visualized:**

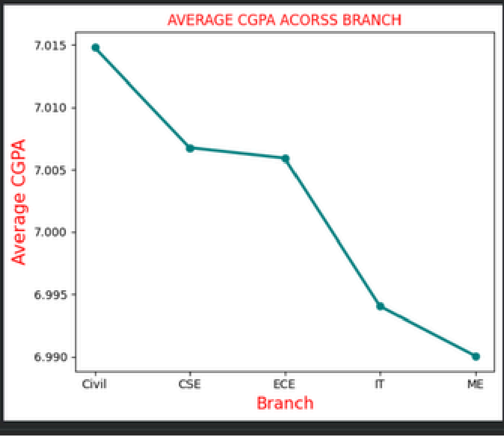
Utilized **Pie Charts and Line Charts** to map the institution's overall placement proportions and trace average CGPA. We used **Histograms and Bar Charts** to look closely at soft skills and see how aptitude test scores compare between placed and unplaced students. We used **Box Plots** to check how much academic backlogs hurt a student's chances

- **Import libraries:** pandas, numpy, matplotlib

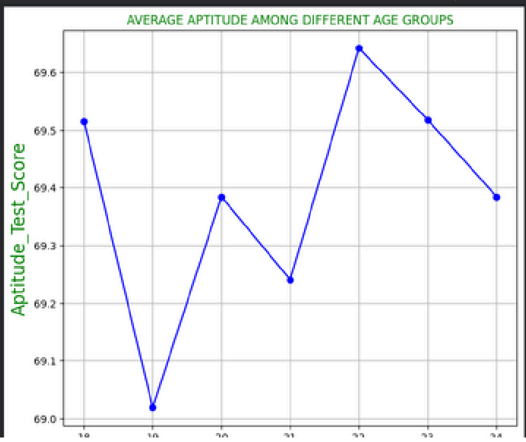
```python
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
```

- **Line chart:**
  - How does the average CGPA vary across departments(Branches)?
  - How does aptitude_test_Score vary across age ?

```python
df=pd.read_csv('/content/Placement-success-dashboard-/train.csv')
avg_cgpa_branch=df.groupby('Branch')['CGPA'].mean().sort_values(ascending=False)
plt.figure(figsize=(6,5))
plt.plot(avg_cgpa_branch.index, avg_cgpa_branch.values, color='teal', marker='o')
plt.xlabel('Branch', fontsize=14, color='red')
plt.ylabel('Average CGPA', fontsize=15, color='red')
plt.title('AVERAGE CGPA ACORSS BRANCH', color='red')
plt.tight_layout()
plt.show()
```

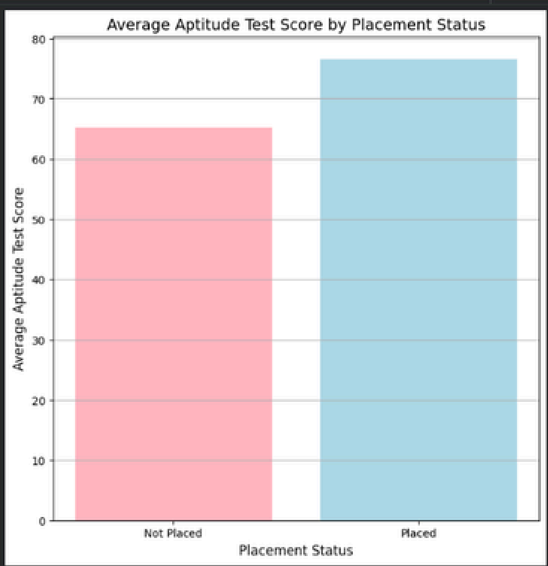



```python
df=pd.read_csv('/content/Placement-success-dashboard-/train.csv')
avg_aptitude_per_age=df.groupby('Age')['Aptitude_Test_Score'].mean()
plt.figure(figsize=(8,7))
plt.plot(avg_aptitude_per_age.index,avg_aptitude_per_age.values,marker='o', color='blue', linewidth='1.5')
plt.xlabel('Age',fontsize=17, color='green')
plt.ylabel('Aptitude_Test_Score', fontsize=17, color='green',)
plt.title('AVERAGE APTITUDE AMONG DIFFERENT AGE GROUPS', color='green')
plt.grid(True)
plt.show()
```



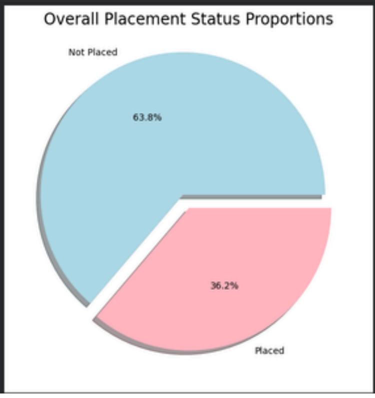
- **Bar chart:** Create a bar chart comparing aptitude_test_score by placement_status?

```python
avg_aptitude_by_placement = df.groupby('Placement_Status')['Aptitude_Test_Score'].mean().reset_index()
plt.figure(figsize=(8, 8))
plt.bar(avg_aptitude_by_placement['Placement_Status'], avg_aptitude_by_placement['Aptitude_Test_Score'], color=['lightpink','lightblue'])
plt.xlabel('Placement Status', fontsize=12)
plt.ylabel('Average Aptitude Test Score', fontsize=12)
plt.title('Average Aptitude Test Score by Placement Status', fontsize=14)
plt.grid(axis='y')
plt.show()
```



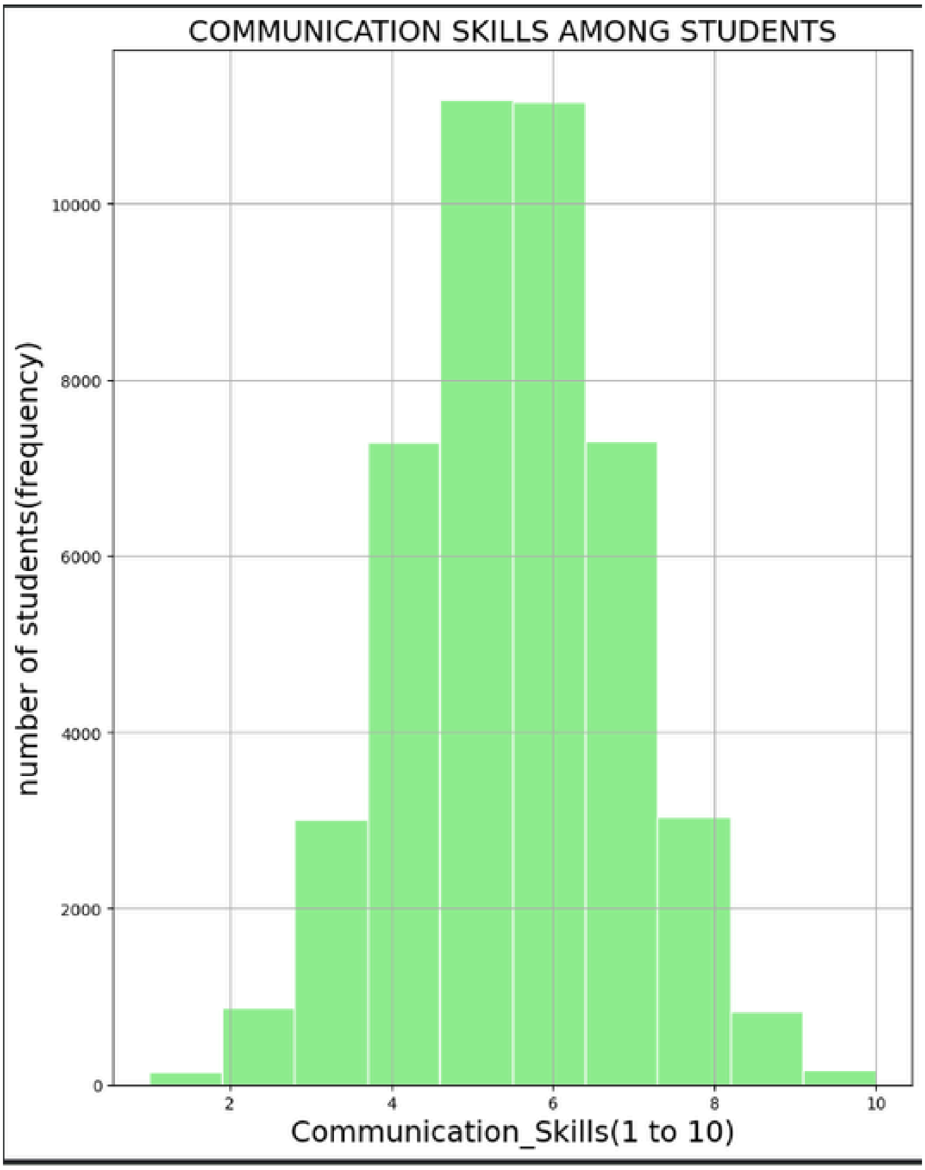
- **Pie chart:** What percentage of students got placed ?

```python
import matplotlib.pyplot as plt
placement_counts = df['Placement_Status'].value_counts()
colors = ['lightblue', 'lightpink']
explode = (0.1, 0.001)
plt.figure(figsize=(7, 8))
plt.pie(sizes, labels=labels, autopct='%1.1f%%', colors=colors, explode=explode, shadow=True)
plt.title('Overall Placement Status Proportions', fontsize=17)
plt.show()
```



- **Histogram:** Distribution of communication skills among the students ?

```python
plt.figure(figsize=(8,10))
plt.hist(df['Communication_Skills'], bins=10, color='lightgreen', edgecolor='white')
plt.xlabel('Communication_Skills(1 to 10)',fontsize=18)
plt.ylabel('number of students(frequency)',fontsize=18)
plt.title('COMMUNICATION SKILLS AMONG STUDENTS', fontsize=18)
plt.grid(True)
plt.tight_layout()
plt.show()
```

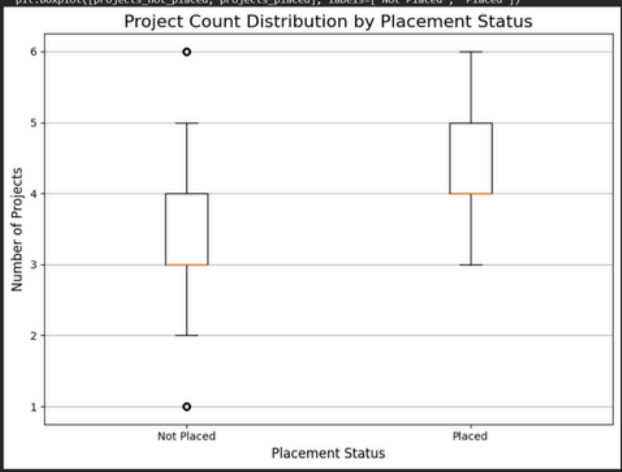


- **Box plot:** How does the total project count differ between placed and non-placed students?

```python
import matplotlib.pyplot as plt
projects_placed = df[df['Placement_Status'] == 'Placed']['Projects']
projects_not_placed = df[df['Placement_Status'] == 'Not Placed']['Projects']

plt.figure(figsize=(8, 6))
plt.boxplot([projects_not_placed, projects_placed], labels=['Not Placed', 'Placed'])
plt.title('Project Count Distribution by Placement Status', fontsize=16)
plt.xlabel('Placement Status', fontsize=12)
plt.ylabel('Number of Projects', fontsize=12)
plt.grid(axis='y')
plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_916/36911878.py:6: MatplotlibDeprecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_labels' since Matplotlib 3.9; support
  plt.boxplot([projects_not_placed, projects_placed], labels=['Not Placed', 'Placed'])
```



3. <u>KPI Analysis :</u>

We used basic Python functions to analyze the dataset. We calculated the total students, placed and not placed students, overall placement rate, CGPA statistics, and average internships and certifications. A pivot table was also used to identify the department with the highest placement success.

- **Total, avg, min, max:**

```python
df['CGPA'].sum()
```

```
np.float64(315103.05999999994)
```

2. **average value** of CGPA()

```python
df['CGPA'].mean()
```

```
np.float64(7.0022902222222205)
```

3. **min and max** values of cgpa()

```python
min_cgpa= df['CGPA'].min()
max_cgpa= df['CGPA'].max()
print(min_cgpa)
print(max_cgpa)
```

```
4.5
9.8
```

- **Total no. of students:**

```
len(df)
print ("total number of students:", len(df))
```

```
total number of students: 45000
```

- **Total no.of placed and non-placed students:**

2. Total **no of placed students** present in the dataset

```
placed_students= df[df['Placement_Status'] == 'Placed']
total_placed_Students = len(placed_students)
print("Total number of placed students:", total_placed_Students)
```

```
Total number of placed students: 16312
```

3. total **no of not placed students** present in the dataset

```
non_placed_students= df[df['Placement_Status'] == 'Not Placed']
total_non_placed_students= len(non_placed_students)
print("Total no of non placed students:", total_non_placed_students)
```

```
Total no of non placed students: 28688
```

- **Overall placement Students:**

4. Total **placement rate** of the dataset

```
placement_rate = (total_placed_Students / total_stu) * 100
print(f"Total placement rate: {placement_rate:.2f}%")
```

```
Total placement rate: 36.25%
```

- **Average internships and certifications:**

6. **Average Internships** of the dataset

```
average_internships= df['Internships'].mean()
print(average_internships)
```

```
0.7740888888888889
```

8. **average certifications** of the dataset

```
avg_certi= df['Certifications'].mean()
print("total no of average certifiations:", avg_certi)
```

```
total no of average certifiations: 1.8009555555555556
```

- **Highest placement rate:**

```python
pivot = df[df['Placement_Status'] == 'Placed'].pivot_table(
    index='Branch',
    values='Student_ID',
    aggfunc='count')
top_dept  = pivot['Student_ID'].idxmax()
top_count = pivot['Student_ID'].max()
print(f"Highest Placement Department : {top_dept} ({top_count} students)")
```

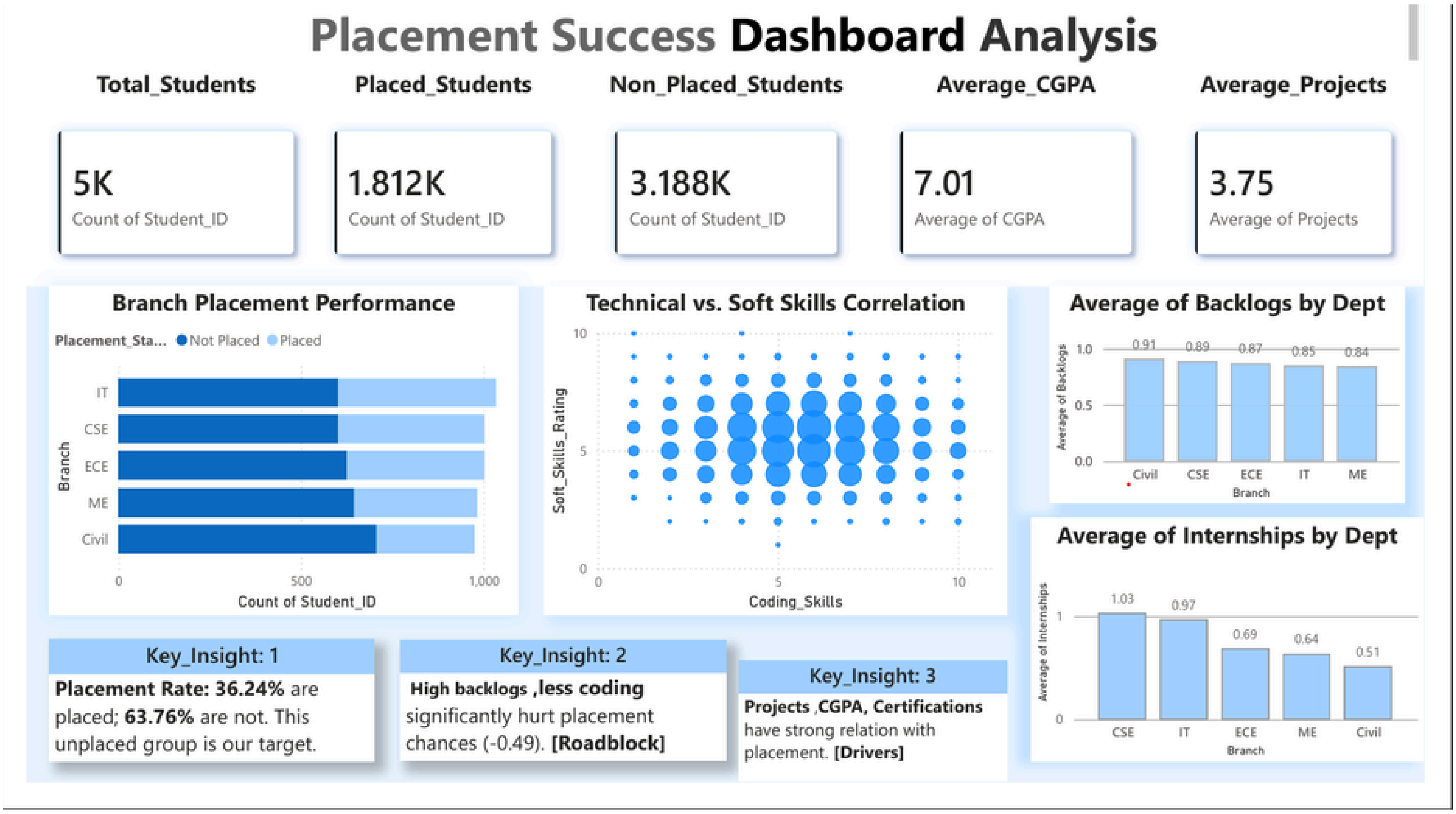
```
Highest Placement Department : CSE (3832 students)
```

```python
pivot = df[df['Placement_Status'] == 'Placed']
pivot.head(7)
```

	Student_ID	Age	Gender	Degree	Branch	CGPA	Internships	Projects	Coding_Skills	Communication_Skills	Aptitude_Test_Score	Soft_Skills_Rating	Certifications	Backlogs	Placem
3	19467	22	Male	MCA	ME	7.78	2	4	6	6	90	4	2	0	
4	23094	20	Female	B.Tech	ME	7.63	1	4	6	5	79	6	2	0	
5	8710	20	Male	BCA	ECE	7.99	1	4	5	7	84	6	2	0	
11	11924	19	Male	MCA	ECE	7.48	2	5	9	8	67	6	3	0	
13	31588	18	Male	B.Tech	ECE	6.74	2	4	7	6	73	8	2	0	
15	6245	18	Male	BCA	ECE	6.79	0	3	5	5	81	6	2	0	
18	25157	23	Male	B.Tech	CSE	6.69	1	4	8	7	68	6	2	1	

4. Dashboard analysis:

Now that our data is cleaned and processed, we imported it into Power BI to look for deep patterns. We split our visual analysis into two distinct dashboards: **Page 1** focuses our **current recruitment problems**, and **Page 2** focuses on **creating a targeted action plan to fix them**.



Dashboard 1: Current situation

- **Dashboard Findings & Analysis:**

1. What does the current placement situation look like?

Business Insight: Out of the **total 5,000 students**, **3,188 students are currently unplaced**, while only **1,812 students have secured placements**. This highlights the need for **targeted training, skill enhancement, and placement preparation** programs to improve student employability.

2. Which branches are struggling the most in placements??

Business Insights: The chart highlights placement performance across departments by comparing placed and unplaced student counts. **Civil and Mechanical engineering show the largest gaps** between placed and unplaced students, indicating **weaker placement outcomes**. where these deparments require training sessions, career guidance and industrial exposure.

3. Do technical coding skills or soft skills matter more for securing a job?

Business Insights: This chart compares students coding skills and soft skills. The analysis shows that students who perform well in **both technical and soft skills have better placement oppurtunities**. Having strong coding skills alone may not be enough, as companies also value communication and teamwork abilities during interviews.

4. What are the biggest challenges students face in different departments?

Business Insights: The analysis shows that **Civil Engineering has the highest average number of backlogs (0.91) and the lowest average internship count (0.51)** among all departments. Students with more backlogs and less practical exposure often face problems during placements. which directly explains their low placement rates

Conclusive point:

This shows that good academic performance like **CGPA helps students** to gain job opportunities during placements. Whereas, Students having active **backlogs affects their placement opportunities.** Because many companies expect students to have zero backlogs. Therefore, **reducing backlog counts should be the main focus area** for improving overall placement_status.

Dashboard 2: Targeted Upskilling and Resource Planning

- ## <u>Dashboard Findings & Analysis:</u>

1. What are the biggest challenges students face in different departments?

Business Insights: The analysis shows that out of 3,188 unplaced students, **Civil Engineering has 709 students** and **Mechanical Engineering has 646 students**. These departments have the largest number of students still looking for jobs. This indicates that they require **"Upskill to hire" program** with industrial exposure and career guidance.

2. What specific skill gaps need to be fixed in the underperforming departments

Business Insights: The analysis shows that communication skills are fairly consistent across all departments. However, **Civil Engineering students have the lowest average coding score (4.48), the fewest projects (3.34), and the lowest number of certifications (1.51).** These skill gaps reduce their placement opportunities and show their lower placement performance compared to other branches.

Conclusive point:

To improve placements, the college should provide training based on the needs of each department. **Civil and Mechanical students should be more encouraged** to **complete more projects and industry certifications** through "upskill _to_hire program" to improve their job opportunities in placements.

- **References:**

- This reference guide was utilized for dashboard design best practices:
  https://www.geeksforgeeks.org/power-bi/power-bi-tutorial/
- This official open-source dataset served as the core foundation for our project.
  https://www.kaggle.com/datasets/sonalshinde123/student-placement-dataset/data